

TARUNKUMAR REDDY THIPPAREDDY

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 3+ years of hands-on coding and architectural leadership. Deep expertise in building and optimizing distributed microservices, designing key features end-to-end, and driving reliability through comprehensive monitoring and observability. Skilled in Java, Spring Boot, React, Kafka, and Cloud Platforms (AWS and Azure). Passionate about AI/ML systems, LLMs, RAG architectures, clean code, and mentoring teams.

EDUCATION

Georgia State University	08/2024 – 08/2025
Master of Science, Computer Science	GPA: 3.7/4.0
Kalasalingam University	06/2018 – 05/2022
Bachelor of Technology, Computer Science and Engineering	GPA: 3.5/4.0

SKILLS

- **Programming Languages:** Java, JavaScript/TypeScript, Python, SQL
- **Backend & APIs:** Spring Boot, Spring Cloud, REST APIs, GraphQL, WebSocket
- **Frontend:** React, Angular, Redux, HTML5, CSS3
- **Data & Messaging:** PostgreSQL, MySQL, MongoDB, Apache Kafka, Redis, RabbitMQ
- **Cloud & Infrastructure:** AWS (EC2, RDS, Lambda, S3, API Gateway, DynamoDB, ECS/EKS, CDK, IAM), Azure
- **DevOps & Tools:** Docker, Kubernetes, Terraform, GitHub Actions, Postman, CI/CD Pipelines, Linux, Jira, Agile, Scrum
- **Testing & Security:** Junit, Mockito, SonarQube, Gatling, OAuth2/OIDC, JWT, Spring Security, API Gateway
- **AI/Agentic:** LLMs, RAG, LangChain, LangGraph, LangSmith, Prompt Engineering

WORK EXPERIENCE

Georgia State University	08/2024 – 08/2025
<i>Application Developer, Assistant</i>	<i>Atlanta, GA, USA</i>
• Built end-to-end AI research assistant using LangChain with a RAG pipeline, handling 10K+ weekly academic queries with 92% accuracy and 30% faster research discovery. • Orchestrated agent workflows with LangGraph and added LangSmith evaluations and guardrails, reducing hallucination incidents by 65% through comprehensive prompt engineering. • Delivered the assistant via AWS Lambda and API Gateway, achieving 99.99% availability with 10% latency reduction through efficient resource allocation. • Developed a Java Spring Boot vector retrieval and citation ranking service using REST and gRPC with PostgreSQL and Redis, deployed on AWS EKS behind API Gateway and Lambda, reduced RAG p95 latency by 28%, scaled 3x, and enabled blue/green deployments.	
DXC Technology	06/2022 – 07/2024
<i>Software Development Engineer</i>	<i>Chennai, India</i>
• Developed scalable microservices and APIs in Java (Spring Boot, Spring Cloud) and Python, implementing robust RESTful and GraphQL endpoints and enabling real-time features with WebSocket streams. • Built responsive single-page applications using modern JavaScript/TypeScript frameworks (React, Angular) with Redux for state management, creating rich, user-friendly interfaces with HTML5 and CSS3. • Designed and optimized data storage with complex SQL for PostgreSQL and MySQL and integrated NoSQL MongoDB. Leveraged messaging and caching systems with Apache Kafka, RabbitMQ, and Redis to enable high-throughput, real-time data pipelines. • Architected and maintained AWS and Azure environments, deploying EC2, ECS/EKS, Lambda, S3, RDS, DynamoDB, and API Gateway for high availability and scale. Established IaC with AWS CDK and Terraform and enforced security with fine-grained IAM roles. • Implemented DevOps best practices with Docker, Kubernetes, and GitHub Actions CI/CD for automated testing and deployment. Streamlined development on Linux using Terraform and managed Agile/Scrum delivery with Jira. • Ensured software quality with automated testing (JUnit, Mockito, Gatling, SonarQube). Strengthened security with OAuth2/OIDC, JWT, Spring Security, and API Gateway policies to safeguard sensitive data.	
Phoenix Softech	06/2021 – 05/2022
<i>Software Development Intern</i>	<i>Chennai, India</i>
• Collaborated with senior engineers to automate financial calculations by developing a Java based backend service integrated with an Angular frontend, and Azure Functions, successfully processing 500k+ records nightly with 100% accuracy. • Optimized SQL queries and introduced connection pooling, improving query performance by 10% using Azure SQL Database indexing strategies. • Achieved over 90% unit-test coverage with JUnit and implemented CI/CD pipelines, ensuring zero critical defects in production deployments.	

CERTIFICATIONS

- Microsoft Certified Azure Developer
- AWS Certified Cloud Practitioner

PROJECTS

Predictive Modelling for Research Trends | *Python, BERT, LSTM*

- Built an **NLP** pipeline with **BERT + LSTM** to forecast research trends, improving prediction accuracy 30%.
- Processed 100k+ abstracts via scalable preprocessing and time-series analysis to support research planning.

Online Learning Management Platform | *React.js, JavaScript, Redux, jsPDF, Axios, Spring Boot, JWT, PostgreSQL*

- Constructed a full-stack Online Learning Platform using ReactJS and Spring Boot, improving course management by 40% and engagement by 30%, supporting 100+ users with seamless remote access and real-time communication.
- Integrated secure **RESTful APIs** with **JWT**-based authentication, providing an engaging and user-friendly educational.

Breast Cancer Health Chatbot | *Golang, Python, Flask, NLP*

- Engineered an **AI-powered** chatbot delivering breast cancer insights, achieving 97% response accuracy, and serving 300 weekly users.

LEADERSHIP & AWARDS

- **Champ Award – DXC:** Led the migration of a monolithic system to cloud-native microservices, achieving 6% improvement in scalability and enabling high availability for critical service.